A

Industrial Oriented Mini Project

On

CYBER THREAT PREDICTIVE ANALYTICS FOR IMPROVING CYBER SUPPLY CHAIN SECURITY

(Submitted in partial fulfilment of the requirements for the award of the Degree)

BACHELOR OF TECHNOLOGY

In

INFORMATION TECHNOLOGY

By

Akula Sahithi (227R1A1265)

Kindigeri Raghu Nath Reddy (227R1A1298)

Batchu Nikhil (227R1A1271)

Under the Guidance of

Mr.M.RAMESH BABU

(Assistant Professor)

DEPARTMENT OF INFORMATION TECHNOLOGY

CMR TECHNICAL CAMPUS

UGC AUTONOMOUS

(Accreditedby NBA& NAAC with'A'Grade Approved By AICTE, NewDelhi and JNTU Hyderabad, Kandlakoya(V), Medchal Road, Hyderabad, Telangana-501401.

2024-2025

DEPARTMENT OF INFORMATION TECHNOLOGY

CERTIFICATE

This is to certify that the project entitled **"Cyber Threat Predictive Analytics for Improving Cyber Supply Chain Security"** has being submitted by **A. Sahithi (227R1A1265), K. Raghu Nath Reddy (217R1A1298), B. Nikhil (217R1A1271),** in partial fulfilment of the requirements for the award of degree of B. Tech in Information Technology of Jawaharlal NehruTechnological University Hyderabad,is a record for the bonafide work carried out by them under our guidance and supervision during the academic year **2024-2025**.

The results embodied in this project have not been submitted by any other university or institute for the award of degree or diploma.

Mr.M.Ramesh Babu
(Assistant Professor)
INTERNAL GUIDE

Dr.B.KavithaRani
HOD

Dr. A. Raji Reddy
DIRECTOR

EXTERNAL EXAMINAR

Submitted for viva voice Examination held on _____

ACKNOWLEDGEMENT

Apart from the efforts of us, the success of any work depends largely on the encouragement and guidelines of many others.We take this opportunity to express our gratitude to the people who have been instrumental in the successful completion of this project.

We take this opportunity to express our profound gratitude and deep regard to our guide faculty, **Mr. M. Ramesh Babu**,Assistant Professor for her exemplary guidance, monitoring and constant encouragement throughout the course of this Project Work.

We take this opportunity to express our profound gratitude to our PRC coordinators, **Ms.G.Menaka,** Assistant Professor and **Mrs.ArfaMahwish**, Assistant Professor. The blessing, help and guidance given by their time to time shall carry us a long way in the journey ahead that, we are about to embark upon.

We are also thankful to **Dr. B. Kavitha Rani,** Professor & Head of the Department for providing excellent infrastructure and a conducive atmosphere for contributing to the successful completion of project.

We obliged to our Director **Dr.A.RajiReddy** for his cooperation throughout the completion of this project.

We wish to express our sincere gratitude to the Management of **CMR Technical Campus**, Hyderabad, **Sri C. Gopal Reddy**, Honorable Chairman, **Smt C. Vasantha Latha**, Honorable Secretary, **Sri C. Abhinav Reddy**, Honorable Chief Executive Officer.

The guidance and support received from all the members of **CMR TECHNICAL CAMPUS** who contributed and who are contributing to this project, was vital for the success of the project. We are grateful for their constant support and help.

Finally, We would like to take this opportunity to thank our family for their constant encouragement without which this assignment would not be possible. We sincerely acknowledge and thank all those who provide support, both directly and indirectly in completion of this project.

A. Sahithi (227R1A1265)
K. Raghu Nath Reddy(227R1A1298)
B. Nikhil (227R1A1271)

ABSTRACT

Cyber Supply Chain (CSC) system is complex which involves different sub-systems performing various tasks. Security in supply chain is challenging due to the inherent vulnerabilities and threats from any part of the system which can be exploited at any point within the supply chain. This can cause a severe disruption on the overall business continuity. Therefore, it is paramount important to understand and predicate the threats so that organization can undertake necessary control measures for the supply chain security. Cyber Threat Intelligence (CTI) provides an intelligence analysis to discover unknown to known threats using various properties including threat actor skill and motivation, Tactics, Techniques, and Procedure (TT and P), and Indicator of Compromise (IoC). This paper aims to analyse and predicate threats to improve cyber supply chain security. We have applied Cyber Threat Intelligence (CTI) with Machine Learning (ML) techniques to analyse and predict the threats based on the CTI properties. That allows to identify the inherent CSC vulnerabilities so that appropriate control actions can be undertaken for the overall cybersecurity improvement. To demonstrate the applicability of our approach, CTI data is gathered and a number of ML algorithms, i.e., Logistic Regression (LG), Support Vector Machine (SVM), Random Forest (RF), and Decision Tree (DT), are used to develop predictive analytics using the Microsoft Malware Prediction dataset. The experiment considers attack and TTP as input parameters and vulnerabilities and Indicators of compromise (IoC) as output parameters. The results relating to the prediction reveal that Spyware/Ransomware and spear phishing are the most predictable threats in CSC. We have also recommended relevant controls to tackle these threats. We advocate using CTI data for the ML predicate model for the overall CSC cyber security improvement.

LIST OF FIGURES

FIGURE NO	NAME OF THE FIGURE	PAGE NO
Figure5.1	Proposed Architecture	13
Figure5.2	Use Case Diagram	15
Figure5.3	Class Diagram	16
Figure5.4	Sequence Diagram	17
Figure5.5	Flowchart Diagram	18
Figure5.6	Dataflow Diagram	19

LIST OF SCREENSHOTS

SCREENSHOT NO	SCREENSHOT NAME	PAGE NO
Screenshot 7.1	User Login	32
Screenshot 7.2	User Registration	32
Screenshot 7.3	View All Remote Users	33
Screenshot 7.4	Trained Test Data	33
Screenshot 7.5	Algorithm Accuracy	34
Screenshot 7.6	View Trained Tested Accuracy Results	34
Screenshot 7.7	View Prediction of Cyber Threat Attack	35
Screenshot 7.8	View Cyber Threat Attack Ratio	35

INDEX

TITLE	PAGENO
ABSTRACT	**i**
LIST OF FIGURES	**ii**
LIST OF SCREEN SHOTS	**iii**
1.INTRODUCTION	**1**
1.1 PROJECT SCOPE	2
1.2 PROJECT PURPOSE	2
1.3 PROJECT FEATURES	3
2.SYSTEM ANALYSIS	**4**
2.1 LITERATURE SURVEY	5
2.2 PROJECT STATEMENT	6
2.3 EXISTING SYSTEM	6
2.4 DISADVANTAGES OF EXISTINGSYSTEM	7
2.5 PROPOSED SYSTEM	7
2.6 ADVANTAGES OF PROPOSED SYSTEM	7
3.SYSTEMSTUDY	**8**
3.1 FEASIBILITY STUDY	9
3.1.1 ECONOMICAL FEASIBILITY	9
3.1.2 TECHNICAL FEASIBILITY	9
3.1.3 SOCIAL FEASIBILITY	9
4.SYSTEM REQUIREMENTS	**10**
4.1 SOFTWARE REQUIREMENTS	11
4.2 HARDWARE REQUIREMENTS	11

5.SYSTEM DESIGN **12**

5.1 INTRODUCTION 13

5.2 ARCHITECTURE 13

5.3 UNIFIED MODELING LANGUAGES 14

5.3.1 USECASE DIAGRAM 15

5.3.2 CLASS DIAGRAM 16

5.3.3 SEQUENCE DIAGRAM 17

5.3.4 FLOWCHART DIAGRAM 18

5.3.5 DATAFLOW DIAGRAM 19

6.IMPLEMENTATION **20**

6.1 MODULES 21

6.2 MODULE DESCRIPTION 21

6.3 SOURCE CODE 22

7.SCREENSHOTS **31**

8.TESTING **36**

8.1 INTRODUCTION TO TESTING 37

8.2 TYPES OF TESTING 37

8.2.1 UNIT TESTING 37

8.2.2 INTEGRATION TESTING 37

8.2.3 FUNCTIONAL TESTING 37

8.2.4 SYSTEM TESTING 38

8.2.5 WHITEBOX TESTING 38

8.2.6 BLACKBOX TESTING 38

8.2.7 PERFORMANCE TESTING 39

8.2.8 ACCEPTANCE TESTING 40

8.2.9 TEST CASES 40

9. CONCLUSION & FUTURESCOPE **41**

9.1 PROJECT CONCLUSION 42

9.2 FUTURE SCOPE 42

10.BIBLIOGRAPHY 44

10.1 REFERENCES 45

10.2 WEBSITES 45

CYBER THREAT PREDICTIVE ANALYTICS FOR IMPROVING CYBER SUPPLY CHAIN SECURITY

1. INTRODUCTION

1. INTRODUCTION

1.1 PROJECT SCOPE

The scope of this project is to enhance the cybersecurity posture of Cyber Supply Chain (CSC) systems, particularly within Smart Cyber-Physical Systems (CPS). These systems are inherently complex and interconnected, which makes them vulnerable to cascading cyberattacks that can originate from a single weak point and spread throughout the system. With many organizations outsourcing parts of their infrastructure to third-party providers, the risk of cyber threats increases significantly. This project focuses on integrating Cyber Threat Intelligence (CTI) and Machine Learning (ML) to detect and predict cyberattack patterns, identify system vulnerabilities, and propose appropriate countermeasures. It aims to tackle sophisticated threats such as Advanced Persistent Threats (APT), command and control operations, and industrial espionage, all of which pose significant risks to CSC environments.

1.2 PROJECT PURPOSE

The primary purpose of this project is to proactively improve the security and resilience of CSC systems by leveraging advanced threat intelligence and predictive analytics. Current approaches often lack the ability to foresee cyber threats based on attacker behavior and historical data. This project addresses that gap by using CTI to gather and analyze information on known attack patterns, threat actor motivations, and tactics. The integration of ML techniques allows for the prediction of potential cyberattacks, enabling organizations to make informed, timely decisions to mitigate threats. By focusing on prediction and prevention, the project aims to shift cybersecurity from a reactive to a proactive approach, ultimately supporting business continuity and reliable service delivery in smart systems.

1.3 PROJECT FEATURES

This project incorporates several key features that distinguish it from existing cybersecurity solutions. First, it integrates CTI and ML in a novel way by mapping threat intelligence attributes— such as Indicators of Compromise (IoCs), Tactics, Techniques, and Procedures (TTPs), and attacker profiles—into classification models. These models, including Logistic Regression, Support Vector Machine, Random Forest, and Decision Tree algorithms, are used to predict the likelihood of cyberattacks. Second, the project uses real-world malware datasets to train and validate the models, ensuring that the predictions are grounded in realistic threat scenarios. The models achieve high accuracy, with results showing up to 85% in threat prediction performance. Third, the project focuses specifically on threats relevant to CSC systems, such as APTs and industrial espionage. Finally, the system provides intelligent recommendations for countermeasures based on the predicted threats, helping organizations implement effective defenses and improve their overall situational awareness.

2.SYSTEM ANALYSIS

2. SYSTEM ANALYSIS

2.1 LITERATURE SURVEY

This research references five key literature surveys related to Cyber Supply Chain (CSC) Security, each offering a distinct perspective on the evolving challenges and solutions within this field.

The first survey, titled **"Cyber Supply Chain Security: Vulnerabilities and Attack Strategies,"** provides a comprehensive examination of the vulnerabilities that exist within cyber supply chains. The survey delves into the implications of these attacks, highlighting their impact on organizational security and the broader cybersecurity ecosystem.

The second survey, **"Integration of Predictive Analytics in Cyber Supply Chain Security,"** explores how predictive analytics can be leveraged to strengthen CSC security. It reviews modern methodologies that employ big data and machine learning to anticipate potential threats before they manifest. The survey evaluates the performance of predictive models in real-world applications and underscores their growing significance in enhancing resilience across digital supply chains.

In the third survey, titled **"Threat Intelligence in Cyber Supply Chains: Current Practices and Future Directions,"** the focus is on the use of cyber threat intelligence (CTI) to secure supply chains. The review assesses current practices in intelligence collection, analysis, and sharing, while identifying critical gaps and limitations.

The fourth survey, **"Case Studies in Cyber Supply Chain Security: Lessons from Notable Attacks,"** analyzes high-profile cyber incidents such as the Dragonfly espionage campaign and the Saudi Aramco power station attack. These real-world insights serve as valuable guidance for organizations aiming to strengthen their cyber supply chain defenses.

Lastly, the fifth survey, titled **"Cyber Supply Chain Risk Management: A Review of Current Frameworks and Challenges,"** reviews various risk management models used in CSC contexts.

2.2 PROJECT STATEMENT

Cyber Supply Chain (CSC) systems are increasingly targeted by sophisticated cyberattacks due to their complexity, interconnectivity, and reliance on third-party service providers. Traditional security measures often fall short in detecting and mitigating these threats, particularly as adversaries exploit vulnerabilities at various nodes in the supply chain. Existing research highlights the growing need for a proactive and intelligent approach to CSC security—one that goes beyond reactive measures and static defenses. This project aims to address this gap by integrating Cyber Threat Intelligence (CTI) and Machine Learning (ML) to enhance the prediction, detection, and prevention of cyber threats in CSC environments. By leveraging threat intelligence attributes such as Indicators of Compromise (IoCs), Tactics, Techniques, and Procedures (TTPs), and attacker motives, and combining them with ML-based classification models, the project seeks to develop a framework capable of identifying and anticipating advanced cyberattacks. This integration is expected to improve situational awareness, support timely decision-making, and ultimately strengthen the overall resilience of CSC systems.

2.3 EXISTING SYSTEM

The existing system for cyber supply chain security primarily focuses on identifying and mitigating vulnerabilities through traditional cybersecurity measures, such as firewalls, intrusion detection systems, and regular security audits. These measures are designed to protect individual components of the supply chain, including hardware, software, and data exchanges. However, the system often lacks a holistic approach that considers the interconnectedness of the supply chain and the potential risks posed by third-party service providers. Organizations typically rely on compliance with industry standards and best practices to secure their supply chains, but this can lead to a false sense of security, as these standards may not be sufficient to address emerging threats. Additionally, while some systems incorporate basic threat detection and incident response mechanisms, Moreover, the existing system tends to focus more on post-incident recovery rather than prevention, leaving supply chains vulnerable to sophisticated cyberattacks. As a result, there is a significant gap in the integration of predictive and intelligence-driven approaches that could enhance overall security and resilience in the face of evolving cyber threats.

2.4 DISADVANTAGES OF EXISTING SYSTEM

1. Lack of Holistic Security
2. Reactive Approach
3. Inadequate Use of Advanced Technologies
4. Over-Reliance on Compliance

2.5 PROPOSED SYSTEM

The proposed system for enhancing cyber supply chain security introduces a more proactive and integrated approach, leveraging advanced technologies like predictive analytics, machine learning, and threat intelligence. Unlike the existing reactive methods, this system emphasizes the early identification and mitigation of potential threats before they can impact the supply chain. By utilizing predictive analytics, the system can analyze vast amounts of data to forecast potential cyber risks, allowing organizations to take preemptive actions to secure vulnerable points in the supply chain. Furthermore, the proposed system adopts a holistic security strategy that considers the entire supply chain network, including third-party service providers. It integrates threat intelligence to provide real-time insights into emerging threats, enabling organizations to adapt quickly to new attack vectors. This intelligence-driven approach enhances the system's ability to detect and respond to sophisticated cyber threats that might bypass traditional security measures. The proposed system also incorporates continuous monitoring and automated response mechanisms to ensure rapid detection and containment of cyber incidents. By focusing on both prevention and real-time response, the system aims to significantly reduce the risk of supply chain disruptions caused by cyberattacks. Ultimately, this proactive and comprehensive approach to cyber supply chain security is designed to improve resilience, minimize risks, and protect critical business operations from evolving cyber threats.

2.6 ADVANTAGES OF PROPOSED SYSTEM

1. Proactive Threat Mitigation
2. Holistic Security Coverage
3. Real-Time Threat Intelligence
4. Enhanced Incident Response

3 SYSTEM STUDY

3. SYSTEM STUDY

3.1 FEASIBILITY STUDY

The feasibility of the project is analyzed in this phase and business proposal is put forth with a very general plan for the project and some cost estimates. During system analysis the feasibility study of the proposed system is to be carried out. This is to ensure that the proposed system is not a burden to the company. For feasibility analysis, some understanding of the major requirements for the system is essential.

Three key considerations involved in the feasibility analysis are:

3.1.1 ECONOMICAL FEASIBILITY

This study is carried out to check the economic impact that the system will have on the organization. The amount of fund that the company can pour into the research and development of the system is limited. The expenditures must be justified. Thus the developed system as well within the budget and this was achieved because most of the technologies used are freely available. Only the customized products had to be purchased.

3.1.2 TECHNICAL FEASIBILITY

This study is carried out to check the technical feasibility, that is, the technical requirements of the system. Any system developed must not have a high demand on the available technical resources. This will lead to high demands on the available technical resources. This will lead to high demands being placed on the client. The developed system must have a modest requirement, as only minimal or null changes are required for implementing this system.

3.1.3 SOCIAL FEASIBILITY

The aspect of study is to check the level of acceptance of the system by the user. This includes the process of training the user to use the system efficiently. The user must not feel threatened by the system, instead must accept it as a necessity. The level of acceptance by the users solely depends on the methods that are employed to educate the user about the system and to make him familiar with it. His level of confidence must be raised so that he is also able to make some constructive criticism, which is welcomed, as he is the final user of the system.

4.SYSTEM REQUIREMENTS

4 SYSTEM REQUIREMENTS

4.1 SOFTWARE REQUIREMENTS

- OperatingSystem Windows 11
- FrontEnd HTML, CSS,JavaScript
- BackEnd Python
- DataBase WampServer2.4
- Framework Django

4.2 HARDWARE REQUIREMENTS

- Processor : IntelCorei5
- HardDisk : 512 GB
- RAM : 8 GB

5 SYSTEM DESIGN

5. SYSTEM DESIGN

5.1 INTRODUCTION

System design, utilizing the Unified Modeling Language (UML), provides a standardized visual language for specifying, visualizing, constructing, and documenting software systems. It integrates best engineering practices to express the design of software projects through graphical notations, crucial for developing object-oriented software.

5.2 ARCHITECTURE

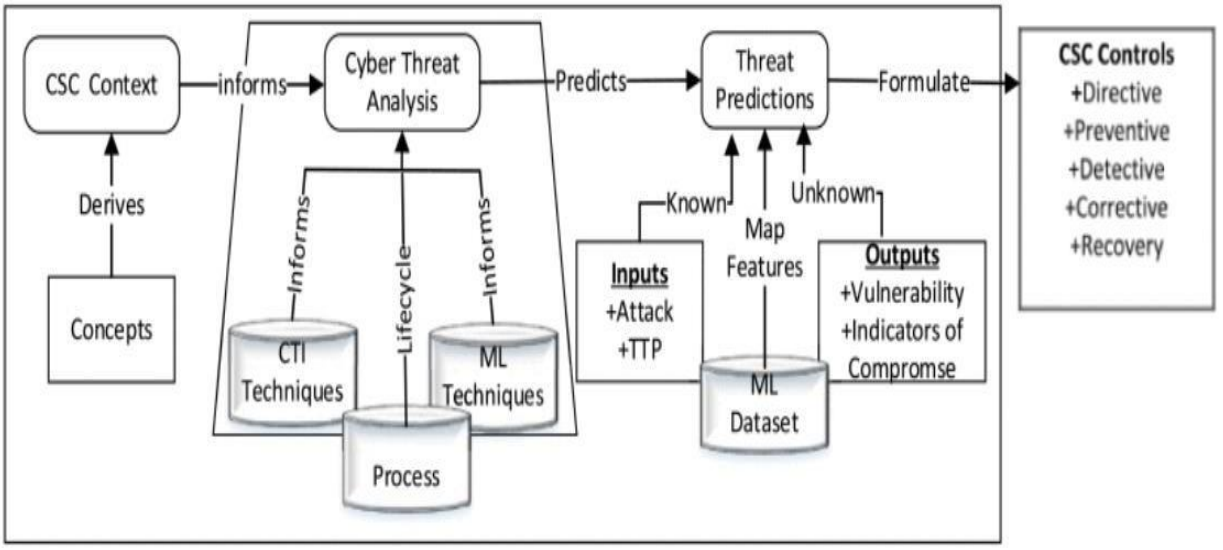


FIGURE 5.1 PROPOSED ARCHITECTURE

5.3 UNIFIED MODELING LANGUAGE(UML)

UML stands for Unified Modelling Language. UML is a standardized general-purpose modelling language in the field of object-oriented software engineering. The standard is managed, and was created by, the Object Management Group. The goal is for UML to become a common language for creating models of object oriented computer software. In its current form UML is comprised of two major components: a Meta-model and a notation. In the future, some form of method or process may also be added to; or associated with, UML. The Unified Modelling Language is a standard language for specifying, Visualization, Constructing and documenting the artifacts of software system, as well as for business modelling and other non-software systems.

The UML represents a collection of best engineering practices that have proven successful in the modelling of large and complex systems. The UML is a very important part of developing objects oriented software and the software development process. The UML uses mostly graphical notations to express the design of software projects.

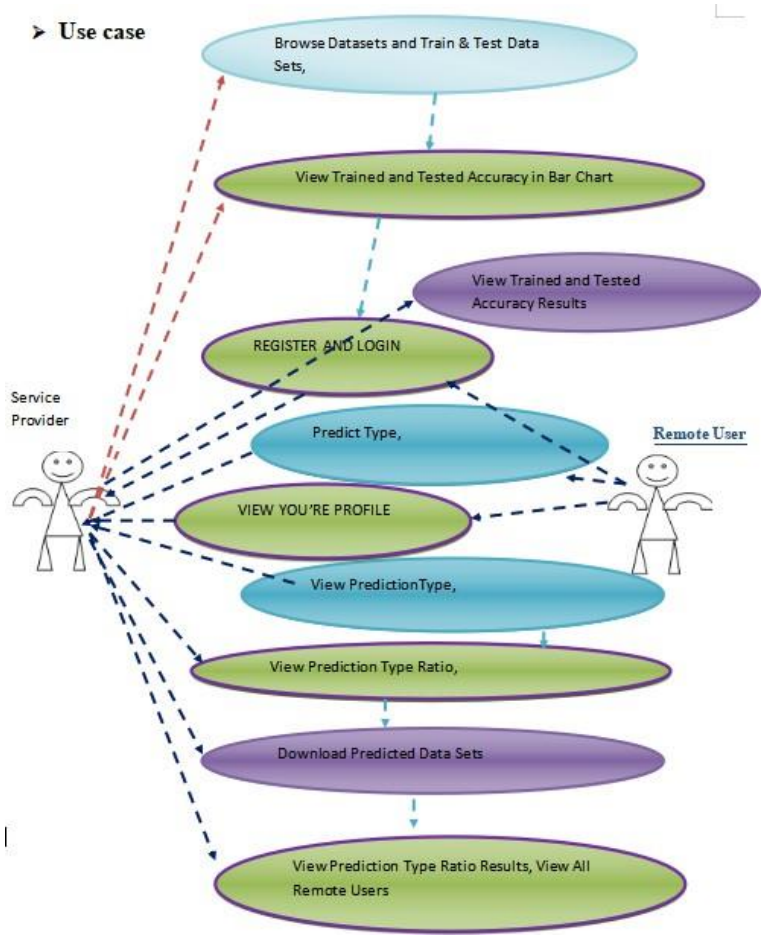
GOALS OF UML:

The Primary goals in the design of the UML are as follows:

1. Provide users a ready-to-use, expressive visual modelling Language so that they can develop and exchange meaningful models.
2. Provide extendibility and specialization mechanisms to extend the core concepts.
3. Be independent of particular programming languages and development process.
4. Provide a formal basis for understanding the modelling language.
5. Encourage the growth of OO tools market.
6. Support higher level development concepts such as collaborations, frameworks, patterns and components.
7. Integrate best practices.

5.3.1 USECASE DIAGRAM

A use case diagram in the Unified Modelling Language (UML) is a type of behavioral diagram defined by and created from a Use-case analysis. Its purpose is to present a graphical overview of the functionality provided by a system in terms of actors, their goals (represented as use cases), and any dependencies between those use cases. The main purpose of a use case diagram is to show what system functions are performed for which



actor. Roles of the actors in the system can be depicted.

FIGURE 5.2 USECASE DIAGRAM

5.3.2 CLASS DIAGRAM

In software engineering, a class diagram in the Unified Modelling Language (UML) is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes, operations(ormethods), and the relationships among the classes. It explains which class contains information.

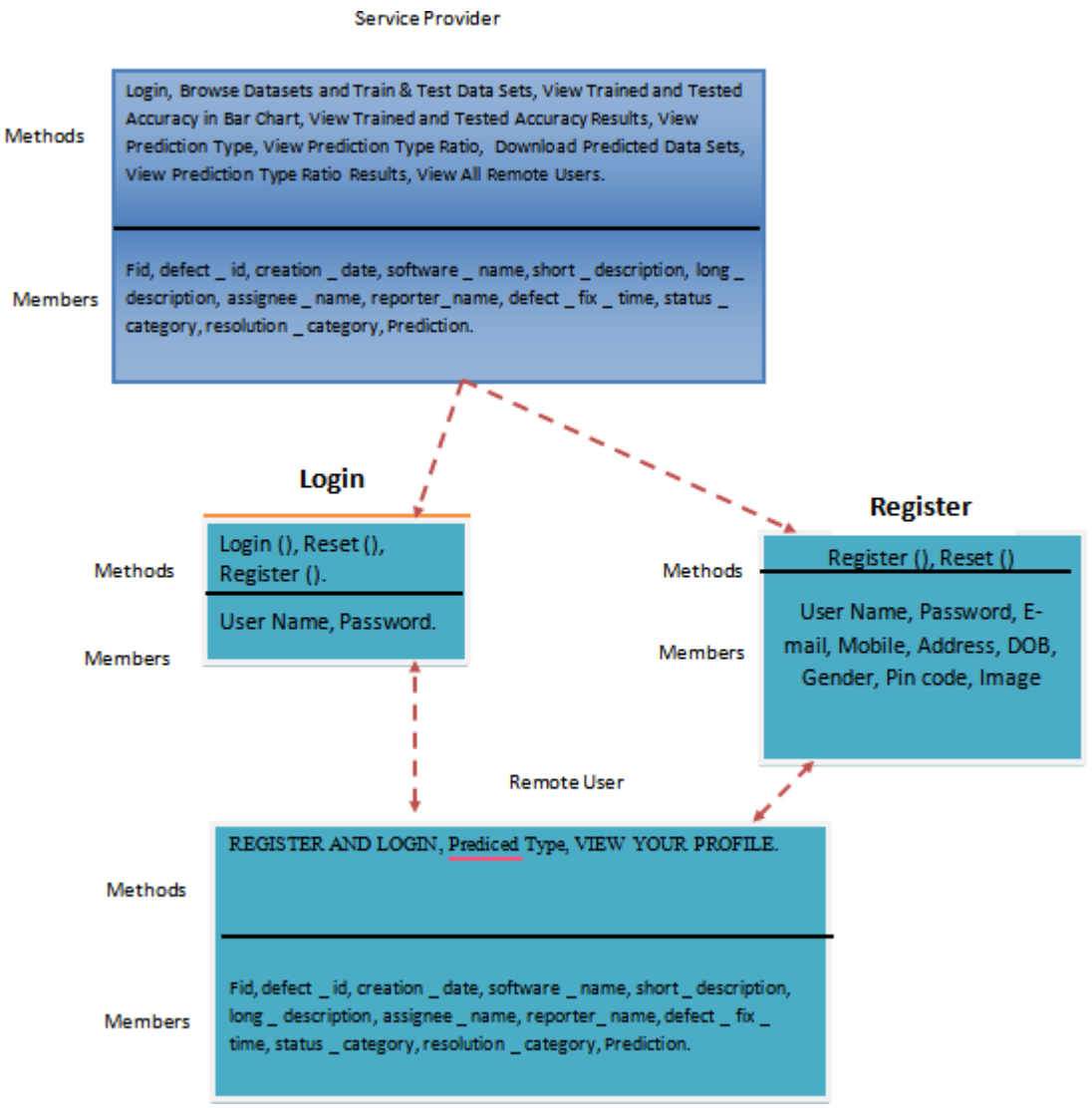


FIGURE 5.3 CLASS DIAGRAM

5.3.3 SEQUENCE DIAGRAM

A sequence diagram in Unified Modelling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. Sequence diagrams are sometimes called event diagrams, event scenarios, and timing diagrams.

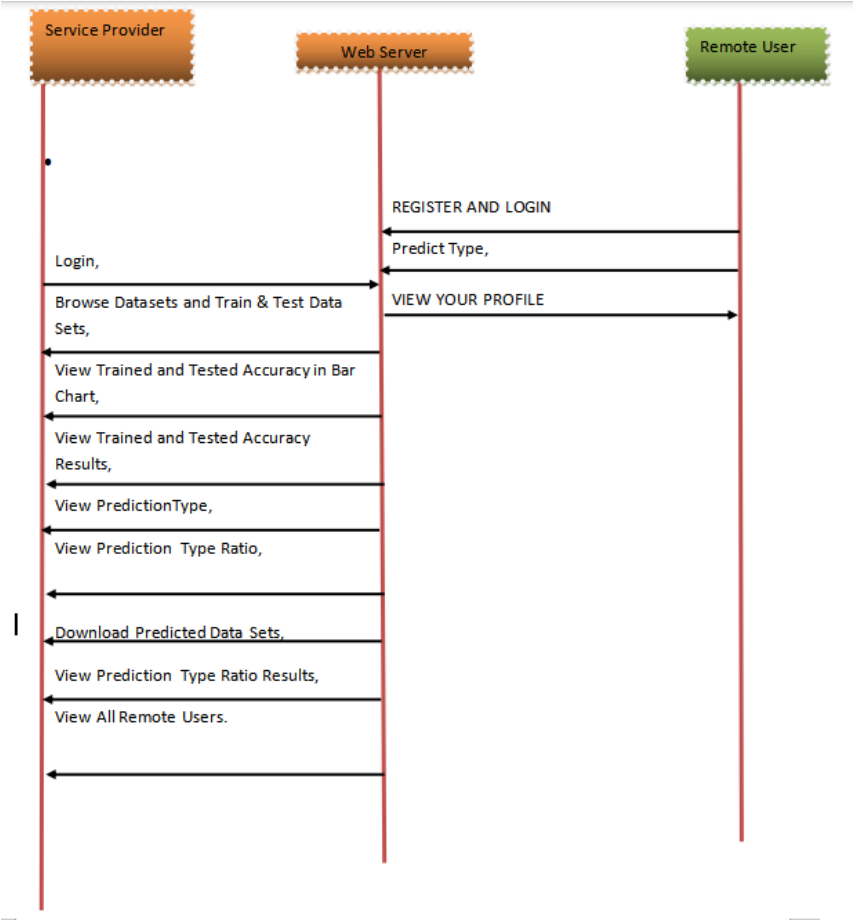
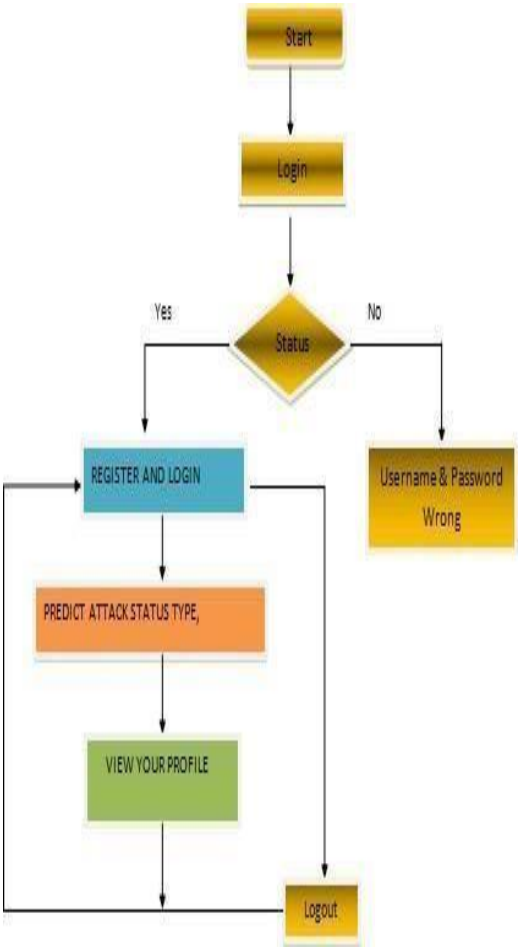
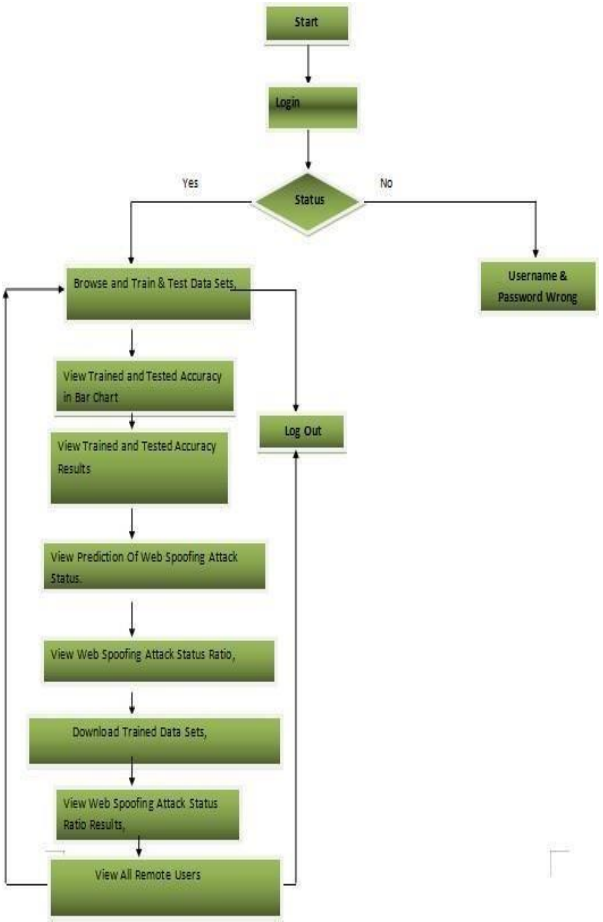


FIGURE 5.4 SEQUENCEDIAGRAM

5.3.3 FLOWCHART DIAGRAM:

A flow chart diagram is a visual tool that shows the steps in a processor work-flow. It uses shapes like arrows and rectangles to represent different actions and decisions. Flow charts help simplify complex tasks and make it easier to understand how things work. They are useful for planning and identifying areas for improvement. Overall, flowcharts help communicate processes clearly to others.





- RemoteUser **- ServiceProvider**

FIGURE 5.5 FLOWCHART DIAGRAM

5.3.4 DATAFLOW DIAGRAM:

A data flow diagram (DFD) is a simple drawing that shows how information moves in a system. It uses arrows to show where data goes, circles to represent processes that use the data, and rectangles for places that store data. DFDs help people see how different parts of a system work together and how data is shared. They are helpful for planning and improving systems by making It clear what data comes in and goes out of each process. Overall, DFDs make it easy to understand how data is handled in a system.

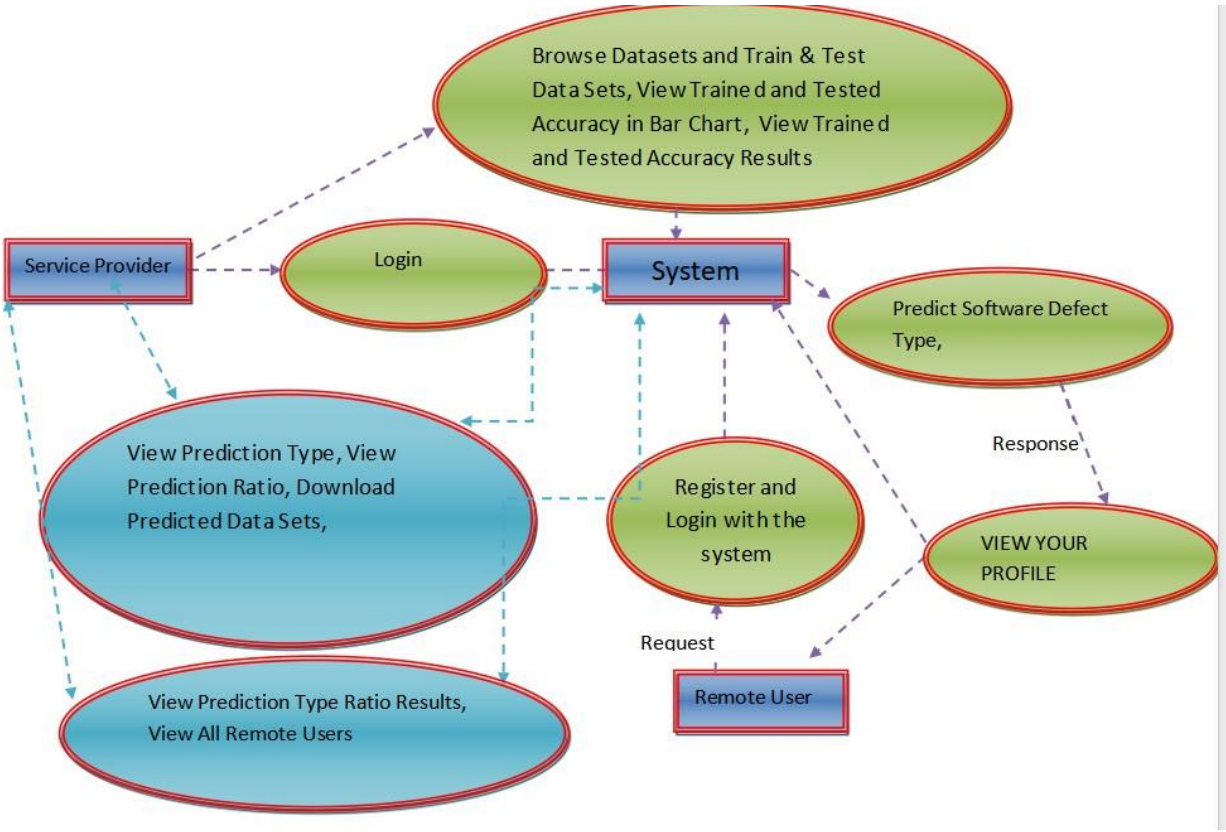


FIGURE: 5.6 DATAFLOW DIAGRAM

6.IMPLEMENTATION

6.1 MODULES

1. Service Provider.
2. View and Authorize Users.
3. Remote User.

6.2 MODULES DESCRIPTION

Service Provider

In this module, the Service Provider is responsible for managing and overseeing the entire system. Their functions include logging in, browsing, and training or testing data sets. They can view trained and tested accuracy results in bar charts and raw formats. The service provider can also view and predict the status of web spoofing attacks, check the web spoofing attack status ratio, and download trained data sets. Additionally, they have the authority to view the spoofing attack status ratio results and monitor all remote users connected to the system.

1. View and Authorize Users

In this module, the admin can View and Authorize Users functionality allows the service provider to monitor and manage remote users. This includes verifying their credentials, reviewing their activities, and ensuring only authorized users have access to sensitive operations, such as downloading or modifying data sets.

2. Remote User

In this module, the Remote User, on the other hand, interacts with the system by registering and logging in. They can predict the attack status type and view their profile details. Their role is limited to using the system's features rather than managing or controlling it.

6.3 SOURCE CODE FOR REMOTE USER:

Views.py

```
from django.db.models import Count
from django.db.models import Q
from django.shortcuts import render, redirect, get_object_or_404
import datetime
import openpyxl
import nltk
import re
import string
from nltk.corpus import stopwords
from sklearn.feature_extraction.text import CountVectorizer
from nltk.stem.wordnet import WordNetLemmatizer
from sklearn.ensemble import RandomForestClassifier
import pandas as pd
from wordcloud import WordCloud, STOPWORDS
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
from sklearn.metrics import accuracy_score
from sklearn.metrics import f1_score
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import VotingClassifier
# Create your views here.
FromRemote_User.modelsimport
ClientRegister_Model,Cyber_model,detection_ratio,detection_accuracy
def login(request):
if request.method == "POST" and 'submit1' in request.POST:
username = request.POST.get('username')
password = request.POST.get('password')
try:
enter = ClientRegister_Model.objects.get(username=username,password=password)
request.session["userid"] = enter.id
return redirect('ViewYourProfile')
except:
pass
return render(request,'RUser/login.html')
def Add_DataSet_Details(request):
return render(request, 'RUser/Add_DataSet_Details.html', {"excel_data": ''})
def Register1(request):
if request.method == "POST":
username = request.POST.get('username')
email = request.POST.get('email')
password = request.POST.get('password')
phoneno = request.POST.get('phoneno')
country = request.POST.get('country')
state = request.POST.get('state')
city = request.POST.get('city')
ClientRegister_Model.objects.create(username=username,email=email,password=password,phon
eno=pho
neno,country=country, state=state, city=city)
return render(request, 'RUser/Register1.html')
```

```
else:
return render(request,'RUser/Register1.html')
def ViewYourProfile(request):
userid = request.session['userid']
obj = ClientRegister_Model.objects.get(id= userid)
return render(request,'RUser/ViewYourProfile.html',{'object':obj})
def Predict_CyberThreat_Type(request):
if request.method == "POST":
review = request.POST.get('keyword')
if request.method == "POST":
Name_of_Covered_Entity= request.POST.get('Name_of_Covered_Entity')
State= request.POST.get('State')
Individuals_Affected= request.POST.get('Individuals_Affected')
Date_of_Breach= request.POST.get('Date_of_Breach')
Location_of_Breached_Information= request.POST.get('Location_of_Breached_Information')
Date_Posted_or_Updated= request.POST.get('Date_Posted_or_Updated')
breach_start= request.POST.get('breach_start')
year= request.POST.get('year')
Source_Ip= request.POST.get('Source_Ip')
Destination_Ip= request.POST.get('Destination_Ip')
data = pd.read_csv("Cyber_Threat.csv")
# data.replace([np.inf, -np.inf], np.nan, inplace=True)
mapping = {'Theft': 0,
'Loss': 1,
'Disclosure': 2,
'Hacking': 3,
'Improper Disposal': 4
}
data['Label'] = data['Type_of_Breach'].map(mapping)
x = data['Name_of_Covered_Entity']
y = data['Label']
# data.drop(['Type_of_Breach'],axis = 1, inplace = True)
cv = CountVectorizer()
print(x)
print(y)
labeled = 'labeled_data.csv'
data.to_csv(labeled, index=False)
data.to_markdown
x = cv.fit_transform(x)
models = []
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(x, y, test_size=0.20)
X_train.shape, X_test.shape, y_train.shape
print("Naive Bayes")
from sklearn.naive_bayes import MultinomialNB
NB = MultinomialNB()
NB.fit(X_train, y_train)
predict_nb = NB.predict(X_test)
naivebayes = accuracy_score(y_test, predict_nb) * 100
print(naivebayes)
print(confusion_matrix(y_test, predict_nb))
print(classification_report(y_test, predict_nb))
models.append(('naive_bayes', NB))
# SVM Model
```

```
print("SVM")
from sklearn import svm
lin_clf = svm.LinearSVC()
lin_clf.fit(X_train, y_train)
predict_svm = lin_clf.predict(X_test)
svm_acc = accuracy_score(y_test, predict_svm) * 100
print(svm_acc)
print("CLASSIFICATION REPORT")
print(classification_report(y_test, predict_svm))
print("CONFUSION MATRIX")
print(confusion_matrix(y_test, predict_svm))
models.append(('svm', lin_clf))
print("Logistic Regression")
from sklearn.linear_model import LogisticRegression
reg = LogisticRegression(random_state=0, solver='lbfgs').fit(X_train, y_train)
y_pred = reg.predict(X_test)
print("ACCURACY")
print(accuracy_score(y_test, y_pred) * 100)
print("CLASSIFICATION REPORT")
print(classification_report(y_test, y_pred))
print("CONFUSION MATRIX")
print(confusion_matrix(y_test, y_pred))
models.append(('logistic', reg))
print("Decision Tree Classifier")
dtc = DecisionTreeClassifier()
dtc.fit(X_train, y_train)
dtcpredict = dtc.predict(X_test)
print("ACCURACY")
print(accuracy_score(y_test, dtcpredict) * 100)
print("CLASSIFICATION REPORT")
print(classification_report(y_test, dtcpredict))
print("CONFUSION MATRIX")
print(confusion_matrix(y_test, dtcpredict))
models.append(('DecisionTreeClassifier', dtc))
print("SGD Classifier")
from sklearn.linear_model import SGDClassifier
sgd_clf = SGDClassifier(loss='hinge', penalty='l2', random_state=0)
sgd_clf.fit(X_train, y_train)
sgdpredict = sgd_clf.predict(X_test)
print("ACCURACY")
print(accuracy_score(y_test, sgdpredict) * 100)
print("CLASSIFICATION REPORT")
print(classification_report(y_test, sgdpredict))
print("CONFUSION MATRIX")
print(confusion_matrix(y_test, sgdpredict))
models.append(('SGDClassifier', sgd_clf))
print("KNeighborsClassifier")
from sklearn.neighbors import KNeighborsClassifier
kn = KNeighborsClassifier()
kn.fit(X_train, y_train)
knpredict = kn.predict(X_test)
print("ACCURACY")
print(accuracy_score(y_test, knpredict) * 100)
print("CLASSIFICATION REPORT")
```

```
print(classification_report(y_test, knpredict))
print("CONFUSION MATRIX")
print(confusion_matrix(y_test, knpredict))
models.append(('KNeighborsClassifier', kn))
classifier = VotingClassifier(models)
classifier.fit(X_train, y_train)
y_pred = classifier.predict(X_test)
data = [Name_of_Covered_Entity]
vector1 = cv.transform(data).toarray()
predict_text = classifier.predict(vector1)
pred = str(predict_text).replace("[", "")
pred1 = pred.replace("]", "")
prediction = int(pred1)
if prediction == 0:
val = 'Theft'
elif prediction == 1:
val = 'Loss'
elif prediction == 2:
val = 'Disclosure'
elif prediction == 3:
val = 'Hacking'
elif prediction == 4:
val = 'Improper Disposal'
print(prediction)
print(val)
Cyber_model.objects.create(Name_of_Covered_Entity=Name_of_Covered_Entity,
State=State,
Individuals_Affected=Individuals_Affected,
Date_of_Breach=Date_of_Breach,
Location_of_Breached_Information=Location_of_Breached_Information,
Date_Posted_or_Updated=Date_Posted_or_Updated,
breach_start=breach_start,
year=year,
Source_Ip=Source_Ip,
Destination_Ip=Destination_Ip,
Prediction=val)
return render(request, 'RUser/Predict_CyberThreat_Type.html',{'objs': val})
return render(request, 'RUser/Predict_CyberThreat_Type.html')
```

VIEWS.PY

```
from django.db.models import Count, Avg
from django.shortcuts import render, redirect
from django.db.models import Count
from django.db.models import Q
import datetime
import xlwt
from django.http import HttpResponse
import numpy as np
import nltk
import re
import string
from nltk.corpus import stopwords
from sklearn.feature_extraction.text import CountVectorizer
from nltk.stem.wordnet import WordNetLemmatizer
import pandas as pd
```

```
from wordcloud import WordCloud, STOPWORDS
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
from sklearn.metrics import accuracy_score
from sklearn.metrics import f1_scorefrom sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import VotingClassifier
from sklearn.ensemble import RandomForestClassifier
# Create your views here.
fromRemote_User.models import
ClientRegister_Model,Cyber_model,detection_ratio,detection_accuracy
def serviceproviderlogin(request):
if request.method == "POST":
admin = request.POST.get('username')
password = request.POST.get('password')
if admin == "Admin" and password =="Admin":
detection_accuracy.objects.all().delete()
return redirect('View_Remote_Users')
return render(request,'SProvider/serviceproviderlogin.html')
def View_CyberThreat_Type_Ratio(request):
detection_ratio.objects.all().delete()
ratio = ""
kword = 'Theft'
print(kword)
obj = Cyber_model.objects.all().filter(Q(Prediction=kword))
obj1 = Cyber_model.objects.all()
count = obj.count();
count1 = obj1.count();
ratio = (count / count1) * 100
if ratio != 0:
detection_ratio.objects.create(names=kword, ratio=ratio)
ratio1 = ""
kword1 = 'Loss'
print(kword1)
obj1 = Cyber_model.objects.all().filter(Q(Prediction=kword1))
obj11 = Cyber_model.objects.all()
count1 = obj1.count();
count11 = obj11.count();
ratio1 = (count1 / count11) * 100
if ratio1 != 0:
detection_ratio.objects.create(names=kword1, ratio=ratio1)
ratio12 = ""
kword12 = 'Disclosure'
print(kword12)
obj12 = Cyber_model.objects.all().filter(Q(Prediction=kword12))
obj112 = Cyber_model.objects.all()
count12 = obj12.count();
count112 = obj112.count();
ratio12 = (count12 / count112) * 100
if ratio12 != 0:
detection_ratio.objects.create(names=kword12, ratio=ratio12)
ratio123 = ""
kword123 = 'Hacking'
print(kword123)
obj123 = Cyber_model.objects.all().filter(Q(Prediction=kword123))
```

```
obj1123 = Cyber_model.objects.all()
count123 = obj123.count();
count1123 = obj1123.count();
ratio123 = (count123 / count1123) * 100
if ratio123 != 0:
detection_ratio.objects.create(names=kword123, ratio=ratio123)
ratio123a = ""
kword123a = 'Improper Disposal'
print(kword123a)
obj123a = Cyber_model.objects.all().filter(Q(Prediction=kword123a))
obj1123a = Cyber_model.objects.all()
count123a = obj123a.count();
count1123a = obj1123a.count();
ratio123a = (count123a / count1123a) * 100
if ratio123a != 0:
detection_ratio.objects.create(names=kword123a, ratio=ratio123a)
obj = detection_ratio.objects.all()
return render(request, 'SProvider/View_CyberThreat_Type_Ratio.html', {'objs': obj})
def View_Remote_Users(request):
obj=ClientRegister_Model.objects.all()
return render(request,'SProvider/View_Remote_Users.html',{'objects':obj})
def ViewTrendings(request):
topic = Cyber_model.objects.values('topics').annotate(dcount=Count('topics')).order_by('-
dcount')
return render(request,'SProvider/ViewTrendings.html',{'objects':topic})
def charts(request,chart_type):
chart1 = detection_ratio.objects.values('names').annotate(dcount=Avg('ratio'))
return render(request,"SProvider/charts.html", {'form':chart1, 'chart_type':chart_type})
def charts1(request,chart_type):
chart1 = detection_accuracy.objects.values('names').annotate(dcount=Avg('ratio'))
return render(request,"SProvider/charts1.html", {'form':chart1, 'chart_type':chart_type})
def View_Prediction_Of_CyberThreat_Type(request):
obj =Cyber_model.objects.all()
return render(request, 'SProvider/View_Prediction_Of_CyberThreat_Type.html', {'list_objects':
obj})
def likeschart(request,like_chart):
charts =detection_accuracy.objects.values('names').annotate(dcount=Avg('ratio'))
return render(request,"SProvider/likeschart.html", {'form':charts, 'like_chart':like_chart})def
Download_Trained_DataSets(request):
response = HttpResponse(content_type='application/ms-excel')
# decide file name
response['Content-Disposition'] = 'attachment; filename="Predicted_Data.xls"'
# creating workbook
wb = xlwt.Workbook(encoding='utf-8')
# adding sheet
ws = wb.add_sheet("sheet1")
# Sheet header, first row
row_num = 0
font_style = xlwt.XFStyle()
# headers are bold
font_style.font.bold = True
# writer = csv.writer(response)
obj = Cyber_model.objects.all()
data = obj # dummy method to fetch data.
```

```
for my_row in data:
row_num = row_num + 1
ws.write(row_num, 0, my_row.Name_of_Covered_Entity, font_style)
ws.write(row_num, 1, my_row.State, font_style)
ws.write(row_num, 2, my_row.Individuals_Affected, font_style)
ws.write(row_num, 3, my_row.Date_of_Breach, font_style)
ws.write(row_num, 4, my_row.Location_of_Breached_Information, font_style)
ws.write(row_num, 5, my_row.Date_Posted_or_Updated, font_style)
ws.write(row_num, 6, my_row.breach_start, font_style)
ws.write(row_num, 7, my_row.year, font_style)
ws.write(row_num, 8, my_row.Source_Ip, font_style)
ws.write(row_num, 9, my_row.Destination_Ip, font_style)
ws.write(row_num, 10, my_row.Prediction, font_style)
wb.save(response)
return response
def train_model(request):
detection_accuracy.objects.all().delete()
data = pd.read_csv("Cyber_Threat.csv")
# data.replace([np.inf, -np.inf], np.nan, inplace=True)
mapping = {'Theft': 0,
'Loss': 1,
'Disclosure': 2,
'Hacking': 3,
'Improper Disposal': 4
}
data['Label'] = data['Type_of_Breach'].map(mapping)
x = data['Name_of_Covered_Entity']
y = data['Label']
# data.drop(['Type_of_Breach'],axis = 1, inplace = True)
cv = CountVectorizer()
print(x)
print(y)
labeled = 'labeled_data.csv'
data.to_csv(labeled, index=False)
data.to_markdown
x = cv.fit_transform(x)
models = []
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(x, y, test_size=0.20)
X_train.shape, X_test.shape, y_train.shape
print("Naive Bayes")
from sklearn.naive_bayes import MultinomialNB
NB = MultinomialNB()
NB.fit(X_train, y_train)
predict_nb = NB.predict(X_test)
naivebayes = accuracy_score(y_test, predict_nb) * 100
print(naivebayes)
print(confusion_matrix(y_test, predict_nb))
print(classification_report(y_test, predict_nb))
models.append(('naive_bayes', NB))
detection_accuracy.objects.create(names="Naive Bayes", ratio=naivebayes)
# SVM Model
print("SVM")
from sklearn import svm
```

```
lin_clf = svm.LinearSVC()
lin_clf.fit(X_train, y_train)
predict_svm = lin_clf.predict(X_test)
svm_acc = accuracy_score(y_test, predict_svm) * 100
print(svm_acc)
print("CLASSIFICATION REPORT")
print(classification_report(y_test, predict_svm))
print("CONFUSION MATRIX")
print(confusion_matrix(y_test, predict_svm))
models.append(('svm', lin_clf))
detection_accuracy.objects.create(names="SVM", ratio=svm_acc)
print("Logistic Regression")
from sklearn.linear_model import LogisticRegression
reg = LogisticRegression(random_state=0, solver='lbfgs').fit(X_train, y_train)
y_pred = reg.predict(X_test)
print("ACCURACY")
print(accuracy_score(y_test, y_pred) * 100)
print("CLASSIFICATION REPORT")
print(classification_report(y_test, y_pred))
print("CONFUSION MATRIX")
print(confusion_matrix(y_test, y_pred))
models.append(('logistic', reg))
detection_accuracy.objects.create(names="Logistic Regression", ratio=accuracy_score(y_test,
y_pred)
* 100)
print("Decision Tree Classifier")
dtc = DecisionTreeClassifier()
dtc.fit(X_train, y_train)
dtcpredict = dtc.predict(X_test)
print("ACCURACY")
print(accuracy_score(y_test, dtcpredict) * 100)
print("CLASSIFICATION REPORT")
print(classification_report(y_test, dtcpredict))
print("CONFUSION MATRIX")
print(confusion_matrix(y_test, dtcpredict))
models.append(('DecisionTreeClassifier', dtc))
detection_accuracy.objects.create(names="Decision Tree Classifier",
ratio=accuracy_score(y_test,
dtcpredict) * 100)
print("SGD Classifier")
from sklearn.linear_model import SGDClassifier
sgd_clf = SGDClassifier(loss='hinge', penalty='l2', random_state=0)
sgd_clf.fit(X_train, y_train)
sgdpredict = sgd_clf.predict(X_test)
print("ACCURACY")
print(accuracy_score(y_test, sgdpredict) * 100)
print("CLASSIFICATION REPORT")
print(classification_report(y_test, sgdpredict))
print("CONFUSION MATRIX")
print(confusion_matrix(y_test, sgdpredict))
models.append(('SGDClassifier', sgd_clf))
detection_accuracy.objects.create(names="SGD Classifier", ratio=accuracy_score(y_test,
sgdpredict) *
100)
```

```
print("KNeighborsClassifier")
from sklearn.neighbors import KNeighborsClassifier
kn = KNeighborsClassifier()
kn.fit(X_train, y_train)
knpredict = kn.predict(X_test)
print("ACCURACY")
print(accuracy_score(y_test, knpredict) * 100)
print("CLASSIFICATION REPORT")
print(classification_report(y_test, knpredict))
print("CONFUSION MATRIX")
print(confusion_matrix(y_test, knpredict))
models.append(('KNeighborsClassifier', kn))
detection_accuracy.objects.create(names="KNeighborsClassifier",ratio=accuracy_score(y_test,
knpredict) * 100)
print("Random Forest Classifier")
rf = RandomForestClassifier()
rf.fit(X_train, y_train)
pred_rfc = rf.predict(X_test)
print("ACCURACY")
print(accuracy_score(y_test, pred_rfc) * 100)
print("CLASSIFICATION REPORT")
print(classification_report(y_test, pred_rfc))
print("CONFUSION MATRIX")
print(confusion_matrix(y_test, pred_rfc))
detection_accuracy.objects.create(names="Random Forest
Classifier",ratio=accuracy_score(y_test,
pred_rfc) * 100)
labeled = 'labeled_data.csv'
data.to_csv(labeled, index=False)
data.to_markdown
obj = detection_accuracy.objects.all()
return render(request,'SProvider/train_model.html', {'objs': obj}
```

7. SCREENSHOTS

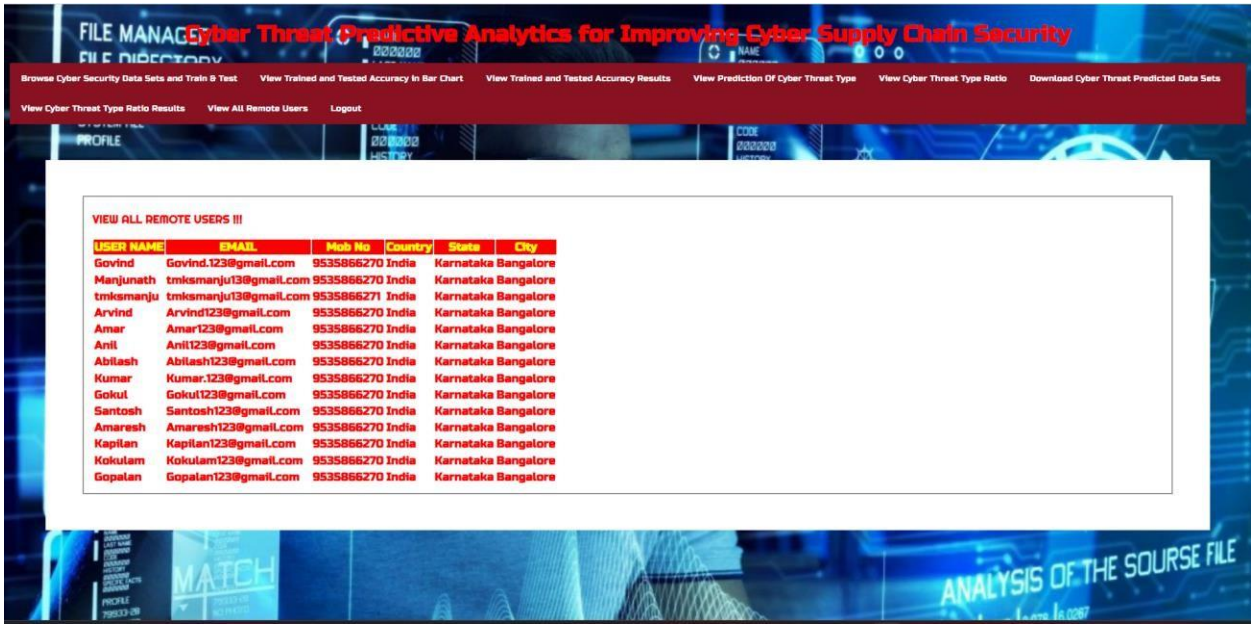
7.1 User Login

Screenshot7.1:UserLogin

7.2 User Registration:

Screenshot7.2:UserRegistration

7.3 View All Remote Users:



USER NAME	EMAIL	Mob No	Country	State	City
Govind	Govind.123@gmail.com	9535866270	India	Karnataka	Bangalore
Manjunath	tmksmanju13@gmail.com	9535866270	India	Karnataka	Bangalore
tmksmanju	tmksmanju13@gmail.com	9535866271	India	Karnataka	Bangalore
Arvind	Arvind123@gmail.com	9535866270	India	Karnataka	Bangalore
Amar	Amar123@gmail.com	9535866270	India	Karnataka	Bangalore
Anil	Anil123@gmail.com	9535866270	India	Karnataka	Bangalore
Abilash	Abilash123@gmail.com	9535866270	India	Karnataka	Bangalore
Kumar	Kumar.123@gmail.com	9535866270	India	Karnataka	Bangalore
Gokul	Gokul123@gmail.com	9535866270	India	Karnataka	Bangalore
Santosh	Santosh123@gmail.com	9535866270	India	Karnataka	Bangalore
Amaresh	Amaresh123@gmail.com	9535866270	India	Karnataka	Bangalore
Kapilan	Kapilan123@gmail.com	9535866270	India	Karnataka	Bangalore
Kokulam	Kokulam123@gmail.com	9535866270	India	Karnataka	Bangalore
Gopalan	Gopalan123@gmail.com	9535866270	India	Karnataka	Bangalore

Screenshot 7.3: RemoteUsers

7.4 Trained Test Data:

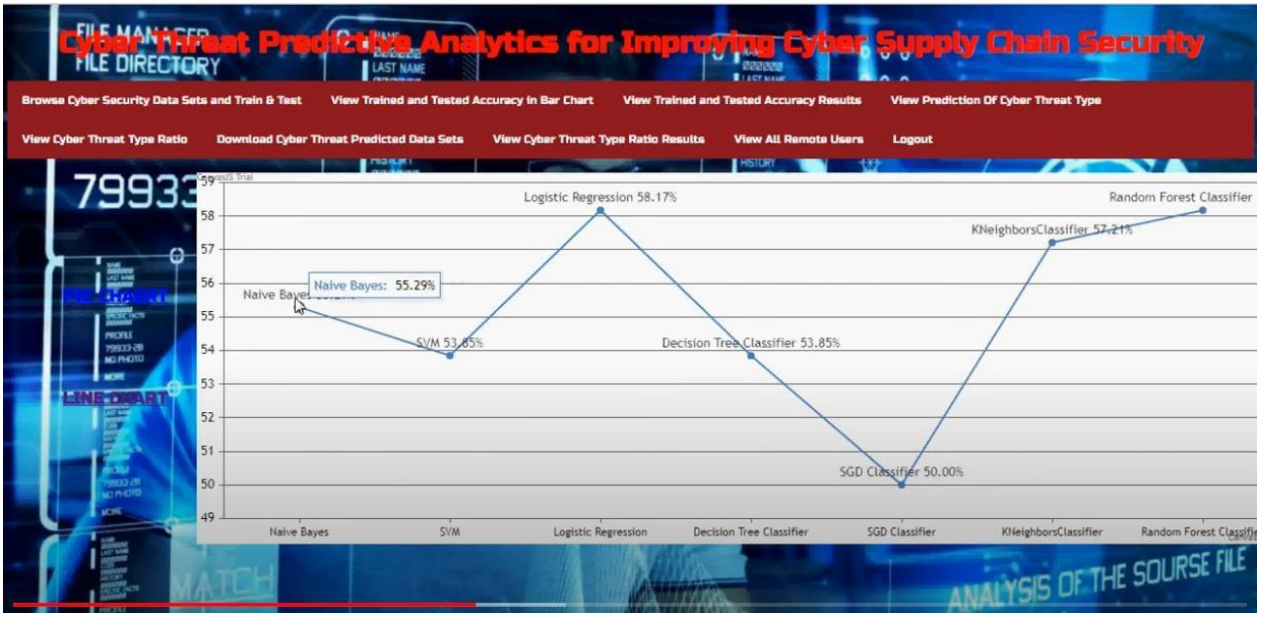
Model Type	Accuracy
Naive Bayes	55.28846153846154
SVM	53.84615384615385
Logistic Regression	58.17307692307693
Decision Tree Classifier	53.84615384615385
SGD Classifier	50.0
KNeighborsClassifier	57.21153846153846
Random Forest Classifier	58.17307692307693

Screenshot7.4: CyberThreatTypeandTestedResults

7.5 Alogrithms Accuracy:

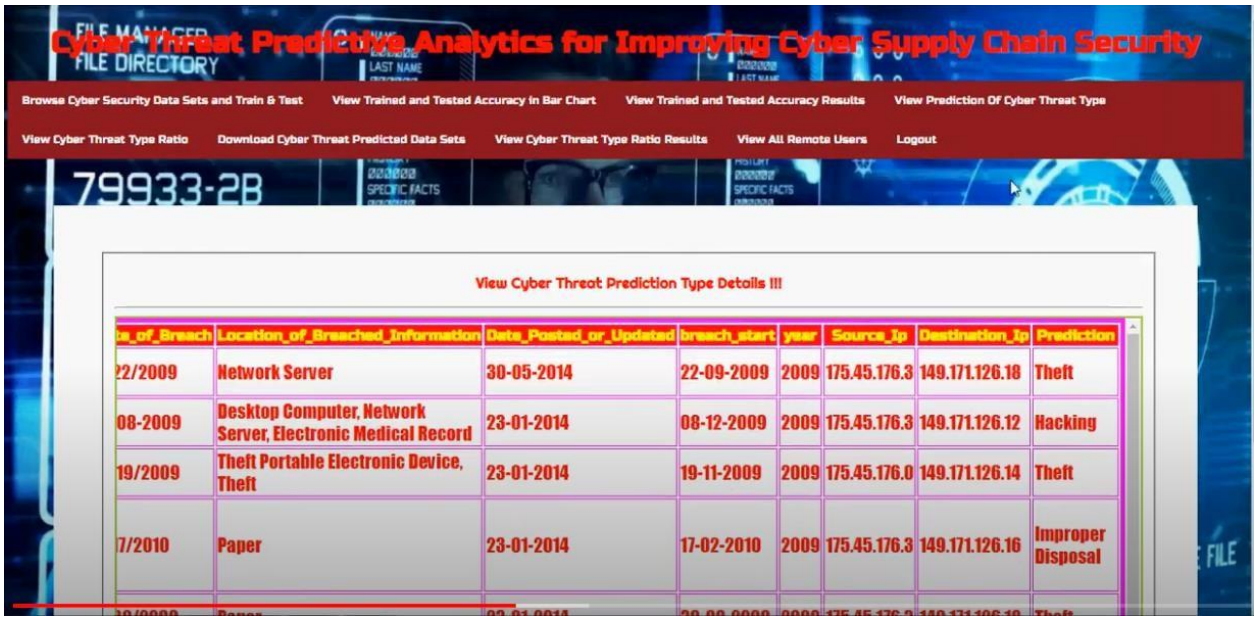
Screenshot7.5: AbovefigureshowsAlgorithmsaccuracybasedonthe given dataset.

7.6 View Trainand Tested Accuracy Results:



Screenshot7.6:ViewTrainandTestedAccuracyResults.

7.7 View Prediction of Cyber Threat Attack:



Screenshot7.7:ViewCyberThreatAttackFoundDetails

7.8 View Cyber Threat Attack Ratio:

Screenshot7.8:ViewCyberThreatAttackFoundRatioDetails

8.TESTING

8.1 INTRODUCTION TO TESTING

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub assemblies, assemblies and/or a finished product It is the process of exercising software with the intent of ensuring that the Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of test. Each test type addresses a specific testing requirement.

8.2 TYPES OF TESTINGS

8.2.1 UNIT TESTING

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program inputs produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application. it is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration.

8.2.2 INTEGRATION TESTING

Integration tests are designed to test integrated software components to determine if they actually run as one program. Testing is event driven and is more concerned with the basic outcome of screens or fields. Integration tests demonstrate that although the components were individually satisfaction, as shown by successfully unit testing, the combination of components is correct and consistent. Integration testing is specifically aimed at exposing the problems that arise from the combination of components.

8.2.3 FUNCTIONAL TESTING

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals.
Functional testing is centered on the following items:

Functions : identified functions must be exercised.

Output : identified classes of application outputs must be exercised. Systems/Procedures : interfacing systems or procedures must be invoked.

Organization and preparation of functional tests is focused on requirements, key functions, or special test cases. In addition, systematic coverage pertaining to identify Business process flows; data fields, predefined processes, and successive processes must be considered for testing. Before functional testing is complete, additional tests are identified and the effective value of current tests is determined.

8.2.4 SYSTEM TESTING

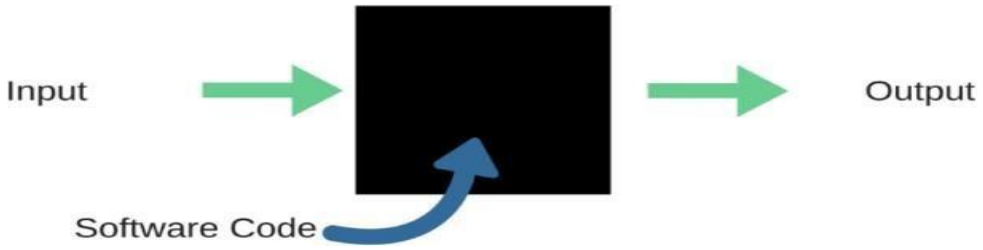
System testing ensures that the entire integrated software system meets requirements. It tests aconfiguration to ensure known and predictable results. An example of system testing is the configuration oriented system integration test. System testing is based on process descriptions and flows, emphasizing pre-driven process links and integration points.

8.2.5 WHITEBOX TESTING

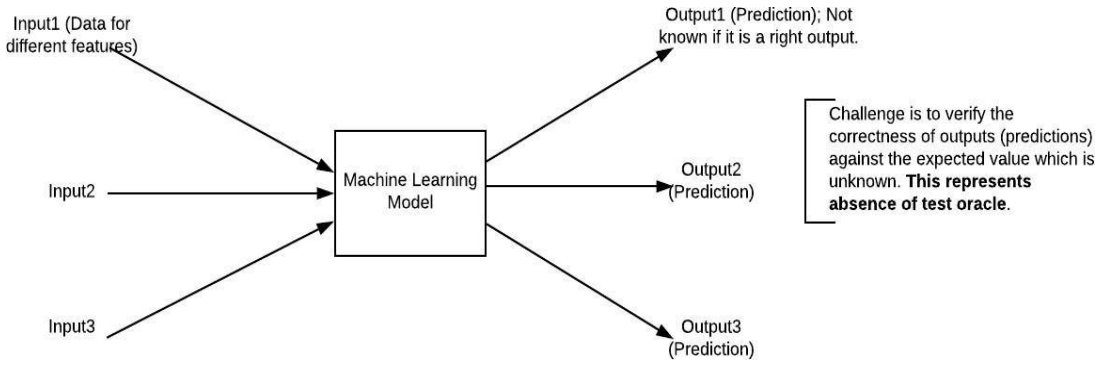
White Box Testing is a testing in which in which the software tester has knowledge of the inner workings, structure and language of the software, or at least its purpose. It is used to test areas that Can not be reached from a black box level.

8.2.6 BLACKBOX TESTING

Black Box Testing is testing the software without any knowledge of the inner workings, structure or language of the module being tested. Black box tests, as most other kinds of tests, must be written from a definitive source document, such as specification or requirements document, such as specification or requirements document. It is a testing in which the software under test is treated, as a black box .you cannot “see” into it. The test provides inputs and responds to outputs without considering how the software works.



When applied to machine learning models, black box testing would mean testing machine learning models without knowing the internal details such as features of the machine learning model,the algorithm used to create the model etc.The challenges,however,is to verify the test outcome against the expected values that are known be forehand.



8.2.7 PERFORMANCE TESTING

Performance testing is a type of software testing that focuses on evaluating the performance and scalability of a system or application. Performance testing aims to identify bottlenecks, measure system performance under various loads and conditions, and ensure that the system can handle the expected number of users or transactions.

8.2.8 ACCEPTANCE TESTING

User Acceptance Testing is a critical phase of any project and requires significant participation by the end user. It also ensures that the system meets the functional requirements.

TestResults: All the test cases mentioned above passed successfully. Nod efects encountered.

8.2.9 TEST CASES

Test Case ID	Test Case Description	Expected Result	Actual Result	Status
TC-001	User login with valid credentials	Login successful	Login successful	Passed
TC-002	User registration with valid details	Registration successful	Registration successful	Passed
TC-003	Predict cyber threat type with valid tweet	Prediction returned correctly	Prediction returned correctly	Passed
TC-004	View user profile after successful login	User details displayed	User details displayed	Passed
TC-005	View prediction of cyber attack status	Status displayed	Status displayed	Passed

9 CONCLUSION&FUTURESCOPE

9.1 PROJECT CONCLUSION

In conclusion, enhancing cyber supply chain security through the integration of advanced machine learning classifiers offers a proactive and comprehensive approach to mitigating the ever-evolving threats that organizations face today. By employing models such as Naive Bayes, SVM, Logistic Regression, Decision Trees, SGD, KNN, and Random Forest, the system can effectively identify, predict, and respond to potential cyber threats before they can cause significant damage.

These classifiers not only improve the accuracy of threat detection but also enable real-time monitoring and rapid response, thereby reducing the risk of disruptions in critical supply chain operations. The implementation of such a system ensures that organizations are better equipped to safeguard their assets, maintain business continuity, and adapt to new challenges in the cyber security landscape. Ultimately, this integrated, intelligence-driven approach represents a significant advancement in the pursuit of robust cyber supply chain security.

9.2 FUTURE SCOPE

1. **Integration of Advanced Threat Intelligence:** Future work should focus on integrating more sophisticated threat intelligence sources into the system.This includes leveraging global threat databases,real-time threat feeds,and collaboration with cybersecurity organizations to enhance the system's ability to detect and respond to emerging threats.

2. **Adaptive Learning Models:** Developing adaptive learning models that can continuously evolve with new threat patterns and attack vectors will be crucial. Implementing techniques such as reinforcement learning or meta-learning could allow the system to dynamically adjust its detection algorithms based on new data and attack trends.

3. **Scalability and Efficiency:** Improving the scalability of the system to handle large-scale, high-velocity data streams from extensive supply chain networks will be essential. This includes optimizing the performance of machine learning models and integrating efficient data processing pipelines to manage the increasing volume and complexity of supply chain data.

4.**Enhanced Privacy and Security Measures:** Future research should also address the privacy and security implications of handling sensitive supply chain data. Implementing privacy-preserving techniques, such as data anonymization and encryption, will be important to protect organizational and customer information while maintaining effective threat detection.

5. **Interoperability and Standardization:** Developing standardized frameworks and protocols for integrating various security tools and systems will improve interoperability. This will facilitate seamless communication between different components of the security infrastructure and enable more cohesive and effective threat management.

6. **User and Behavior Analytics:** Incorporating advanced user and behavior analytics to better understand normal and anomalous behaviors within the supply chain. This could involve analyzing user interactions, transaction patterns, and communication flows to identify subtle indicators of potential threats.

7. **Collaborative Threat Intelligence Sharing:** Promoting collaborative efforts and information sharing among organizations, industry groups, and governmental bodies will enhance collective threat awareness. This approach will help in building more comprehensive threat profiles and improving overall cybersecurity resilience.

10.BIBLIOGRAPHY

10.1 REFERENCES

National Cyber Security Centre. (2018). Example of Supply Chain Attacks. 2. A. Yeboah-Ofori and S. Islam, ‘‘Cyber security threat modelling for supply chain organizational environments,’’ MDPI. Future Internet, vol. 11, no. 3, p. 63, Mar. 2019. 3. B. Woods and A. Bochman, ‘‘Supply chain in the software era,’’ in Scowcroft Center for Strategic and Security. Washington, DC, USA: Atlantic Council, May 2018.

Exploring the Opportunities and Limitations of Current Threat Intelligence Platforms, Version 1, ENISA, Dec. 2017. 5. C. Doerr, TU Delft CTI Labs. (2018). Cyber Threat Intelligences Standards—A High Level Overview. 6. Research Prediction. (2019). Microsoft Malware Prediction. 7. A.Yeboah-Ofori and F. Katsriku, ‘‘Cybercrime and risks for cyber physical systems,’’ Int. J. Cyber-Secur. Digit. Forensics, vol. 8, no. 1, pp. 43–57, 2019. 8. CAPEC-437, Supply Chain. (Oct. 2018). Common Attack Pattern Enumeration and Classification: Domain of Attack. 9. Open Web Application Security Project (OWASP). (2017). The Ten Most Critical Application Security Risks, Creative Commons Attribution-Share Alike 4.0 International License. 10. US-Cert. (2020). Building Security in Software & Supply Chain Assurance.

10.2 WEBSITES

https://repository.uel.ac.uk/item/89q03#:~:text=This%20paper%20aims%20to%20analyse%20and%20predicate%20threats,predict%20the%20threats%20based%20on%20the%20CTI%20properties.

https://ieeexplore.ieee.org/document/9448097

https://www.researchgate.net/publication/352238278_Cyber_Threat_Predictive_Analytics_for_Improving_Cyber_Supply_Chain_Security

https://www.jetir.org/view?paper=JETIRGL06008